

Solutions TP2

1 Résumé des solutions

Exercice 1

```
— sub_info : CameraInfo, '/camera/camera/color/camera_info'  
— sub_rgb : Image, '/camera/camera/color/image_raw'  
— sub_depth : Image, '/camera/camera/depth/image_rect_raw'  
— pub_image : Image  
— pub_points : PointStamped
```

Exercice 2

```
self.fx = msg.k[0]  
self.fy = msg.k[4]  
self.cx = msg.k[2]  
self.cy = msg.k[5]
```

Exercice 3

```
Z = np.median(valid) / 1000.0
```

Exercice 4

```
X = (u - self.cx) * Z / self.fx  
Y = (v - self.cy) * Z / self.fy
```

Exercice 5

```
largeur_m = (x2 - x1) * Z / self.fx  
hauteur_m = (y2 - y1) * Z / self.fy
```

Exercice 6

```
pt = PointStamped()  
pt.header = rgb_msg.header  
pt.point.x = X  
pt.point.y = Y  
pt.point.z = Z  
self.pub_points.publish(pt)
```